

Yuquan Xu

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EDUCATION

National University of Singapore MSc in Precision Health and Medicine Grade: 4.38/5.00	Aug 2025 – Jul 2027 (Expected)
Chengdu University (Joint Undergraduate Programme) BSc (Hons) in Data Science-First Class, University of Stirling Grade: 77.64%, Rank:1/99	Sep 2021 – Jul 2025
BEng in Data Science and Big Data Technology, Chengdu University Grade: 85.84%, Rank: 3/99	

PROFESSIONAL EXPERIENCE

University of California, Irvine Junior Specialist, Department of Radiological Sciences Supervised by Prof. Sabee Molloy	Sep 2026 – Present
– Research in medical image analysis, currently including semi-supervised segmentation and emerging work in dynamic and 4D imaging, with contributions to ongoing collaborative projects in the group.	
Chengdu University Undergraduate Researcher, Stirling College Supervised by Prof. Yuefei Wang	Nov 2022 – Jul 2025
– Conducted research on deep learning for medical image segmentation, contributing to method development, dataset preparation, experimental evaluation, and manuscript preparation.	

PUBLICATIONS

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- [1] Y. Wan, Z. Chen, **Y. Xu**, Y. Xu, M. Li, and Y. Wang, “[Breaking error coupling via divergent-convergent coordination for semi-supervised medical image segmentation](#),” *Medical Image Analysis*, vol. 114, Art. no. 104236, 2026.
 - [2] **Y. Xu**, T. M. Rajeh, Y. Zhang, L. Zhang, Y. Wan, J. Joo, and Y. Wang, “[Enhancing cross-dataset generalization with quasi-multimodal training and the diamond hybrid backbone](#),” *Biomedical Signal Processing and Control*, vol. 119, Art. no. 109902, 2026.
 - [3] X. Wang, Y. Wang, **Y. Xu**, Y. Zhang, and L. Zhang, “[MMTU-Net: Enhancing medical image semantic segmentation with multi-level multi-scale fusion and transformer](#),” *The Visual Computer*, vol. 41, no. 11, pp. 8975–9003, 2025.
 - [4] R. Feng, Y. Wang, J. Xue, **Y. Xu**, Y. Zhang, and X. Yu, “[CLAC-Net: A composite medical image segmentation framework using self-attention and cross-layer asymmetric connections](#),” *The Visual Computer*, vol. 41, no. 6, pp. 3925–3955, 2025.
 - [5] Y. Wang, Y. Zhang, L. Zhang, Y. Wan, Z. Chen, **Y. Xu**, R. Cao, L. Zhao, Y. Yang, and X. Yu, “A feature

enhancement network based on image partitioning in a multi-branch encoder-decoder architecture,” *Knowledge-Based Systems*, vol. 311, Art. no. 113120, 2025.

- [6] Y. Wang, **Y. Xu**, X. Yu, and R. Feng, “Flattened and simplified SSCU-Net: Exploring the convolution potential for medical image segmentation,” *The Journal of Supercomputing*, vol. 80, no. 16, pp. 23471–23518, 2024.
- [7] Y. Wang, Y. Zhang, L. Zhang, **Y. Xu**, R. Feng, H. Cai, J. Xue, Z. Zhao, X. Guo, Y. Wei, Z. Wang, S. Qiu, Y. Yang, and X. Yu, “Multi-Bottleneck progressive propulsion network for medical image semantic segmentation with integrated macro-micro dual-stage feature enhancement and refinement,” *Expert Systems with Applications*, vol. 252, Art. no. 124179, 2024.
- [8] Y. Wang, X. Yu, Y. Yang, S. Zeng, **Y. Xu**, and R. Feng, “FTUNet: A feature-enhanced network for medical image segmentation based on the combination of U-shaped network and vision transformer,” *Neural Processing Letters*, vol. 56, no. 2, Art. no. 83, 2024.

MANUSCRIPTS

- [1] **Y. Xu**, Y. Wan, D. Black, W. Li, Y. Li, and S. Molloy, “From Context to Focus: Uncertainty-Guided Selective Processing for Semi-Supervised Medical Image Segmentation,” manuscript in preparation.
- [2] **Y. Xu**, Y. Zhang, Y. Wan, Y. Yang, C. Xie, J. Song, and Y. Wang, “SGL-Net: Multi-Granularity Attention and Dual-Path Feature Resampling for Medical Image Segmentation,” *Computers and Electrical Engineering*, under review.
- [3] J. Song, Z. Tao, X. Chen, and **Y. Xu**, “Hallucination Severity Is Associated With Interface-versus-External Expression Along a Normative Structural Hierarchy,” *Schizophrenia Bulletin*, under review.

INTERNSHIPS AND ADDITIONAL EXPERIENCE

CASCO Signal Ltd.	May 2025 – Jul 2025
Software Development Engineer Intern	
– Contributed to backend development for the WebCTC railway control system, implementing data-query APIs and supporting business-logic modules.	

HONORS AND AWARDS

Outstanding Graduate of Sichuan Province, Class of 2025	2025
Outstanding Graduation Thesis, Chengdu University	2025
Outstanding Graduate of Chengdu University	2025
First Prize Scholarship, Chengdu University	2023, 2024
Special Prize Scholarship, Chengdu University	2022